Module 3
SCTP DSAI DS2F Coaching

Moneyball Analytics

NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE | Centre for Professional and Continuing Education

SkillsFuture Career Transition Programme (SCTP)

Advanced Professional Certificate in

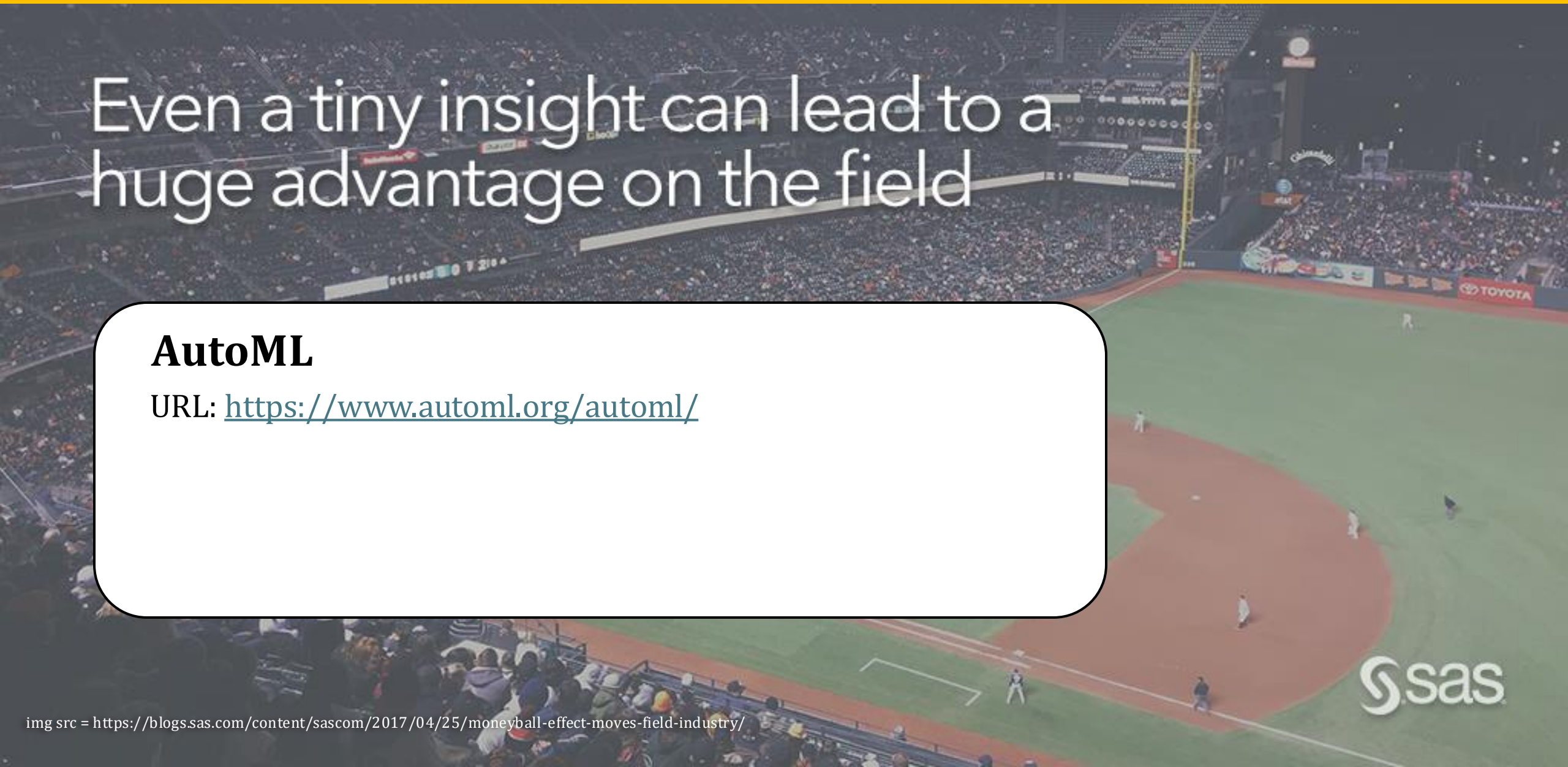
Data Science and AI

Get the Skills, Tools and Mindset to Kickstart Your Tech Career

Moneyball Analytics

img src = https://blogs.sas.com/content/sascom/2017/04/25/moneyball-effect-moves-field-industry/

Moneyball Analytics



AutoML

URL: https://www.automl.org/automl/

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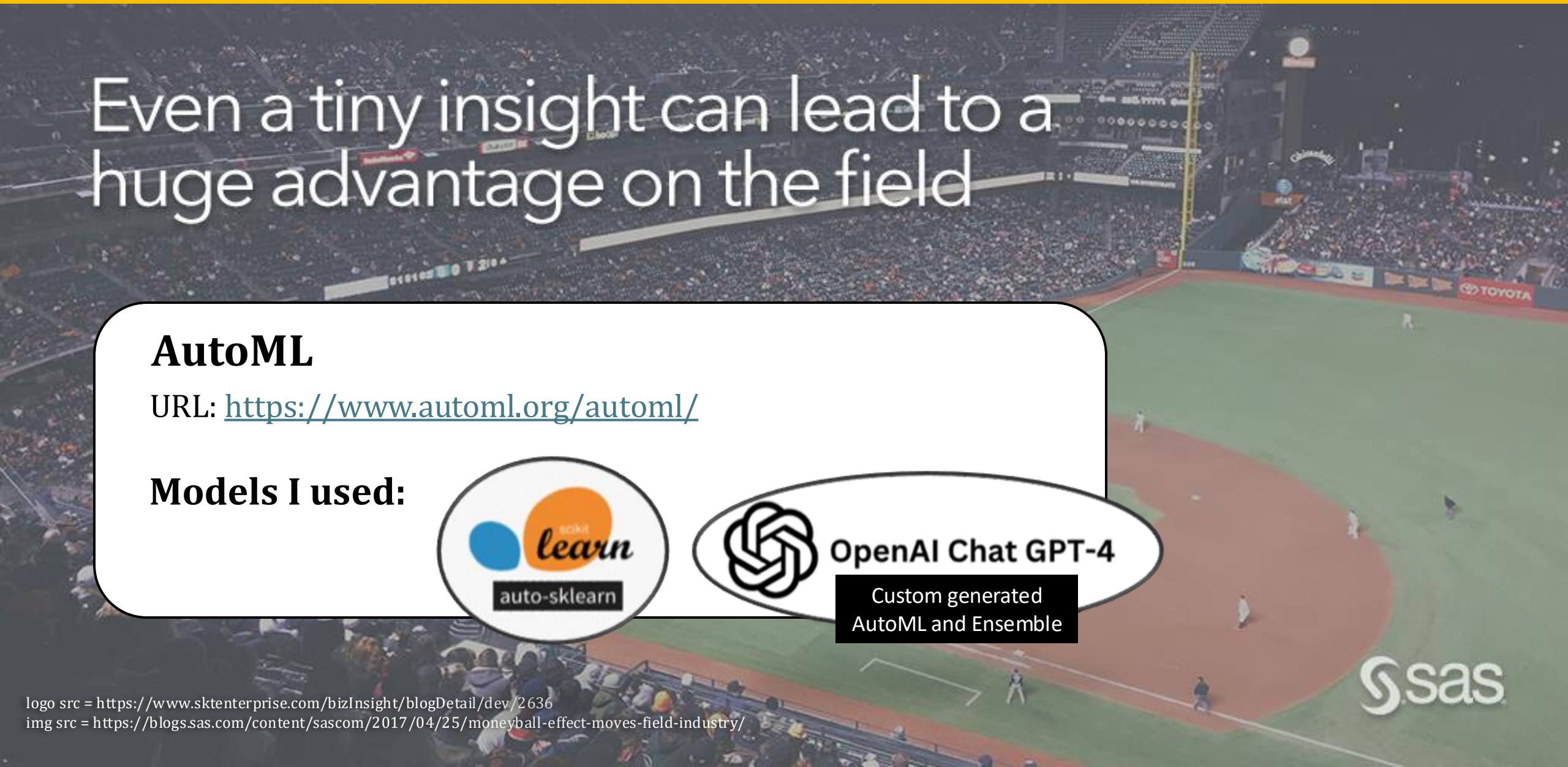
Moneyball Analytics



AutoML

URL: https://www.automl.org/automl/

Models I used:

scikit learn
auto-sklearn

OpenAI Chat GPT-4

Custom generated AutoML and Ensemble

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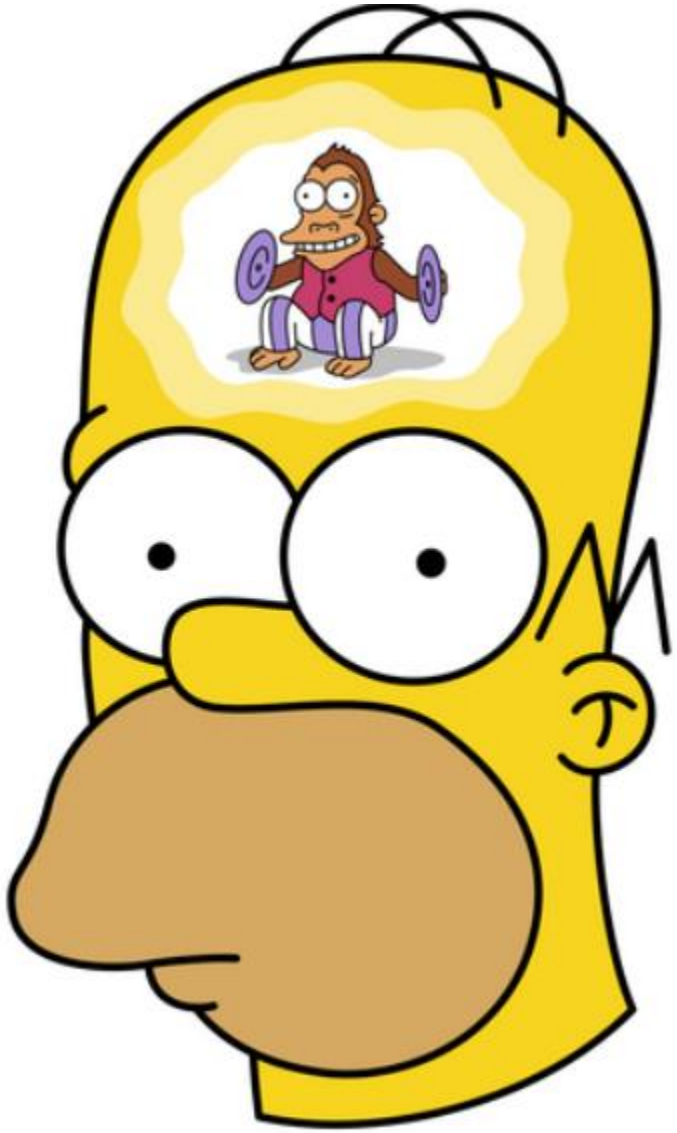
img src = https://blogs.sas.com/content/sascom/2017/04/25/moneyball-effect-moves-field-industry/

Moneyball Analytics

My Challenges & Solutions



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img src = https://mystickermania.com/sticker-packs/simpsons-stickers/simpsons-homer-monkey-head

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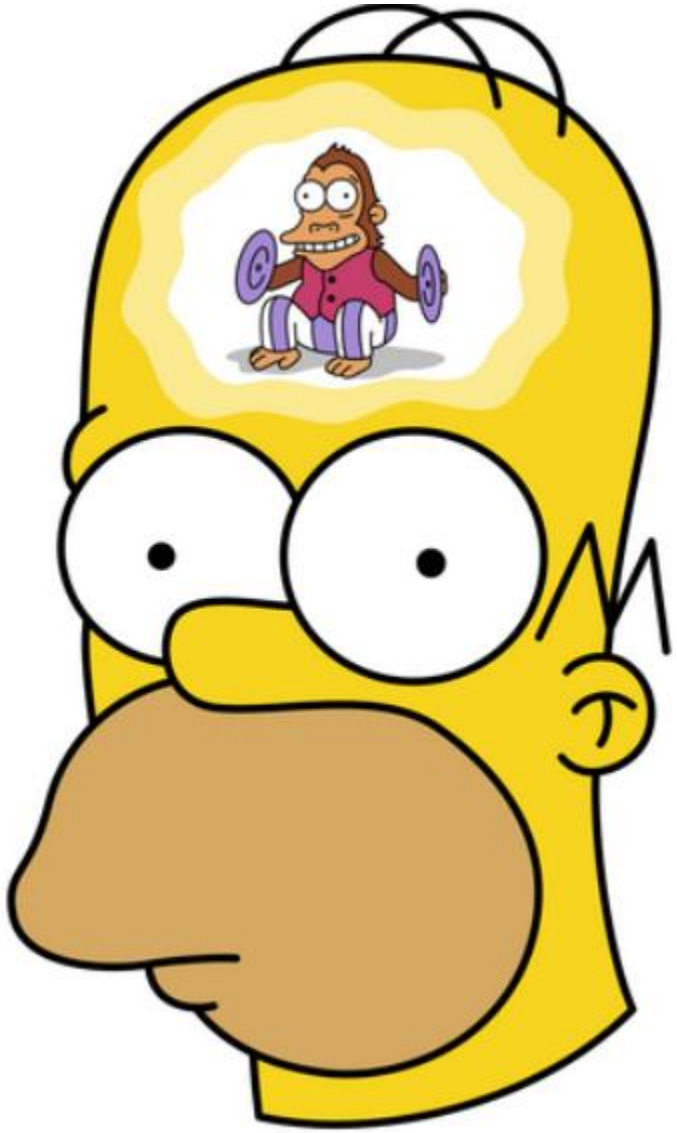
My Challenges & Solutions

Challenge – Variety of Machine Learning models



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My Challenges & Solutions

Challenge – Variety of Machine Learning models

Solution – Use diagrams

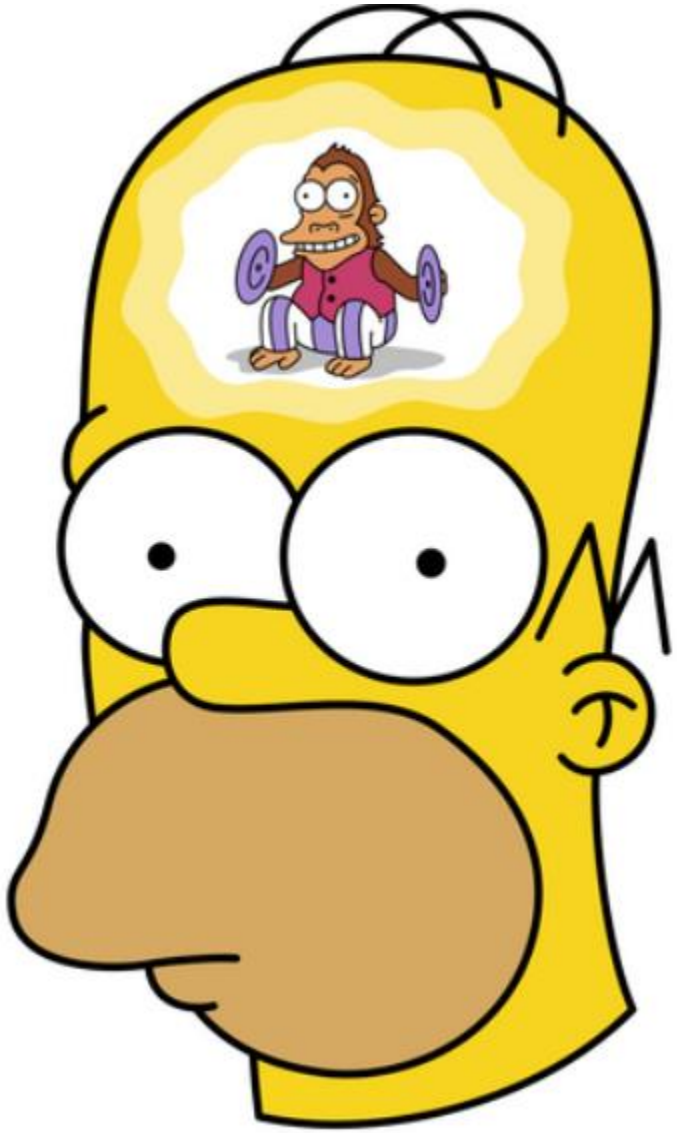
```
START
│
├── 1. What is your task?
│   │
│   ├── Regression (predict number)
│   │   │
│   │   ├── Linear relationship?
│   │   │   ├── YES → Linear Regression / Ridge / Lasso
│   │   │   └── NO  → Decision Tree / Random Forest (shallow)
│   │   │
│   │   └── Very small data (<100 samples)?
│   │       ├── YES → Prefer Linear / Bayesian Regression
│   │       └── NO  → Add Gradient Boosting cautiously
│   │
│   └── Classification (predict category)
│       │
│       ├── Data is linearly separable?
│       │   ├── YES → Logistic Regression / Linear SVM
│       │   └── NO  → Tree-based / Kernel SVM
│       │
│       ├── Few samples (<100)?
│       │   ├── YES → KNN / Naive Bayes / Logistic Regression
│       │   └── NO  → Random Forest / Gradient Boosting
│       │
│       └── High-dimensional data?
│           ├── YES → Naive Bayes / Linear SVM
│           └── NO  → Tree-based models
│
└── 2. Additional constraints
    │
    ├── Need interpretability?
    │   → Linear Models / Decision Trees
    │
    ├── Risk of overfitting high?
    │   → Regularization (Ridge, Lasso)
    │
    ├── Very noisy data?
    │   → Random Forest / Bayesian methods
    │
    └── Extremely small data (<50)?
        → Avoid deep learning entirely
        → Prefer simple + strong priors
```

Moneyball Analytics

My Challenges & Solutions

Challenge – Variety of Machine Learning models
Solution – Use diagrams

Challenge – Lost in math and algorithm



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My Challenges & Solutio

Challenge – Variety of Machin
Solution – Use diagrams

Challenge – Lost in math and
Solution – Use AI to simplify

5.5 Choosing the learning paradigm

5.5.1 Supervised learning

Use when labeled outcomes exist.

Examples: - classification: no-show vs attended - regression: demand quantity - ranking: order candidates by relevance

5.5.2 Unsupervised learning

Use when labels do not exist but structure matters.

Examples: - clustering patients by visit behavior - anomaly detection on claims - dimensionality reduction for visualization or denoising

5.5.3 Reinforcement learning

Use when actions influence future states and rewards over time.

Examples: - dynamic pricing policies - inventory control - treatment recommendation with sequential feedback - robotic control

A practical caution: many business problems that appear to be reinforcement learning are actually **supervised learning on historical outcomes**, because the action-feedback loop is not safely or reliably available.

5.6 Minimum-quality standard

A project should have, before modeling: - data inventory - row-grain definition - label source documentation - data dictionary - train/validation/test split plan - leakage checklist - permissions and privacy review

Moneyball Analytics

My Challenges & Solutions

Challenge – Variety of Machine Learning models
Solution – Use diagrams

Challenge – Lost in math and algorithm
Solution – Use AI to simplify

```
# ============================================================
# MODEL 2 CONFIGURATION — ADJUST THESE HYPERPARAMETERS
# ============================================================

MODEL2_N_TRIALS = 50
MODEL2_CV_SPLITS = 10
MODEL2_USE_GROUP_CV = False
MODEL2_USE_GROUP_KFOLD = True
MODEL2_GROUP_COLUMN = "yearID"
MODEL2_RANDOM_SEED = 42
MODEL2_OUTPUT_FILE = "prediction_submission_2.csv"

# Allowed algorithm families in the random search pool.
MODEL2_CANDIDATE_POOL = [
    "ridge",
    "lasso",
    "elasticnet",
    "random_forest",
    "extra_trees",
    "gradient_boosting",
    "hist_gradient_boosting",
    "lightgbm",
]
```

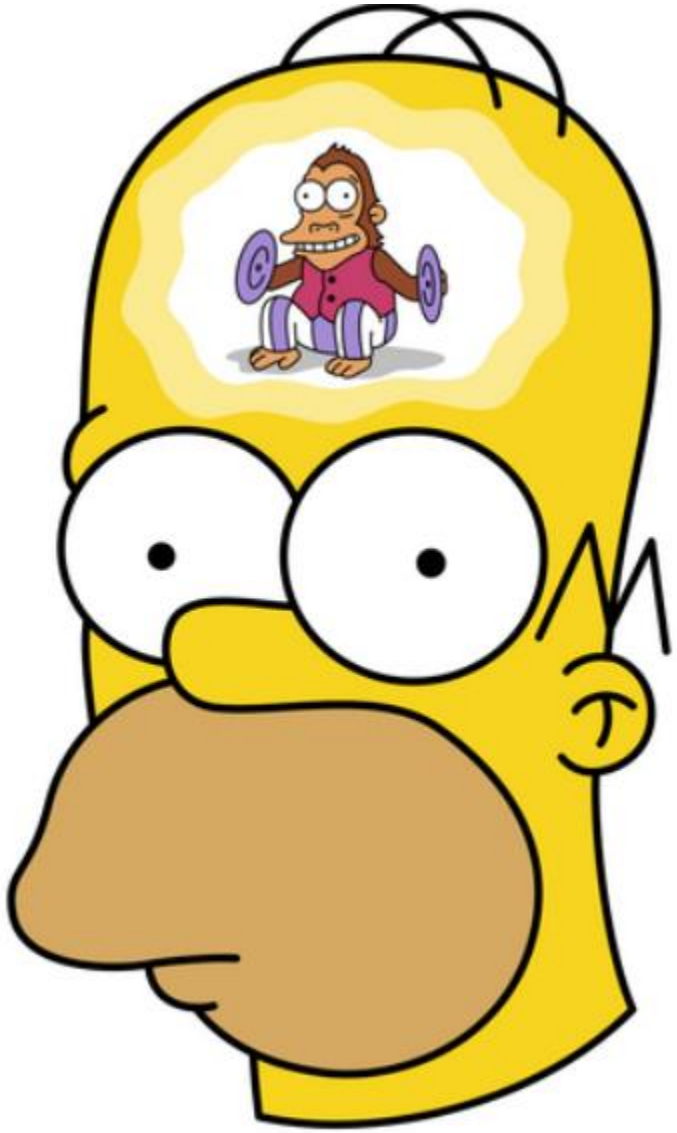
Moneyball Analytics

My Challenges & Solutions

Challenge – Variety of Machine Learning models
Solution – Use diagrams

Challenge – Lost in math and algorithm
Solution – Use AI to simplify

Challenge – Domain knowledge (MLB metrics)
Solution – NotebookLM
(URL: https://notebooklm.google.com/notebook/d7ccb014-dbcb-488a-ab59-4fba63ef5863)



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Model Selection & Findings

Hyper-Param \ # of runs	1	2	3	4	5
TIME_LEFT_FOR_THIS_TASK	1800	2700	3600	5400	7200
PER_RUN_TIME_LIMIT	120	120	120	120	180
MEMORY_LIMIT_MB	4096	4096	4096	4096	4096
ENSEMBLE_SIZE	25	25	50	50	25
RESAMPLING_FOLDS	5	5	5	5	5
N_JOBS	4	4	4	4	4
PRIVATE SCORE (TRUNC)	**3.30**	**3.13**	**3.86**	**3.16**	**3.14**

Custom generated AutoML

Hyper-Param \ # of runs	1	2	3	4	5
N_TRIALS	18	25	30	50	80
CV_SPLITS	5	5	10	10	8
USE_GROUP_CV	True	True	True	False	True
USE_GROUP_KFOLD			True	True	False
GROUP_COLUMN		yearID	yearID	yearID	
RANDOM_SEED	42	42	42	42	42
BEST ALGORITHM	**Lasso**	**Lasso**	**Lasso**	**Lasso**	**Lasso**
PRIVATE SCORE (TRUNC)	**3.16**	**3.16**	**3.16**	**3.29**	**3.36**

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Moneyball Analytics

Model Selection & Findings

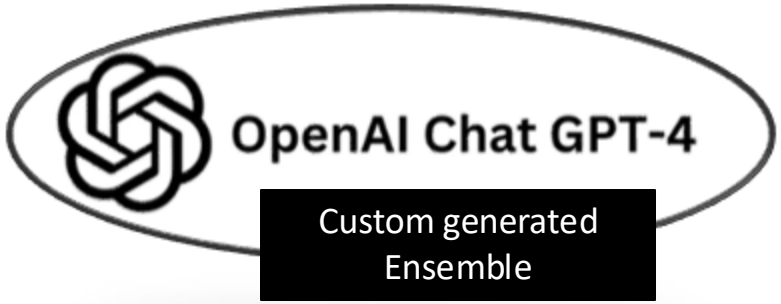
OpenAI Chat GPT-4

Custom generated Ensemble

Hyper-Param \ # of runs	1	2	3	4	5
TOTAL_TRIALS	24	30	35	50	80
TOP_POOL_SIZE	8	10	5	10	15
SELECT_TOP_K	5	5	5		
CV_SPLITS	5	5	5	10	8
USE_GROUP_CV	True	True	True	True	True
ENSEMBLE_SIZE	5	4	3	3	5
GROUP_COLUMN	yearID	yearID	yearID		
RANDOM_STATE	42	42	42		
RANDOM_SEED	123	123	123	123	123
WEIGHTING_METHOD	Inverse MAE				
ENSEMBLE SELECTION	**Ridge 4 ElasticNet**	**Ridge 3 ElasticNet**	**Ridge 3**	**Ridge 3**	**Ridge 4 ElasticNet**
PRIVATE SCORE (TRUNC)	**3.22**	**3.21**	**3.20**	**3.24**	**3.22**

Moneyball Analytics

Model Selection & Findings



Best Personal Private Score

- 3.13 (Auto-SKLearn)

Top Team 5 Private Scores

- 2.99176 (Ridge) by Lewis
- 3.00000 (Ridge) by Lewis

Hyper-Param \ # of runs	1	2	3	4	5
TOTAL_TRIALS	24	30	35	50	80
TOP_POOL_SIZE	8	10	5	10	15
SELECT_TOP_K	5	5	5		
CV_SPLITS	5	5	5	10	8
USE_GROUP_CV	True	True	True	True	True
ENSEMBLE_SIZE	5	4	3	3	5
GROUP_COLUMN	yearID	yearID	yearID		
RANDOM_STATE	42	42	42		
RANDOM_SEED	123	123	123	123	123
WEIGHTING_METHOD	Inverse MAE				
ENSEMBLE SELECTION	**Ridge 4 ElasticNet**	**Ridge 3 ElasticNet**	**Ridge 3**	**Ridge 3**	**Ridge 4 ElasticNet**
PRIVATE SCORE (TRUNC)	**3.22**	**3.21**	**3.20**	**3.24**	**3.22**

Moneyball Analytics

ACKNOWLEDGEMENTS

Portions of the technical design, debugging support, and documentation refinement for this project were assisted by ChatGPT (OpenAI).

All final implementation decisions, data engineering work, analytical modeling, and system integration were performed by the project authors.

Moneyball Analytics – Appendix

NotebookLM Prompt

Domain Research

Using official, reliable and reputable sources only, research US Major League Baseball statistics, including aggregate statistics such as:
i) batting average
ii) on-base percentage
iii) slugging
iv) on-base plus slugging, streaks, etc
since the 1900s until present day. The query should provide deep insight into what makes a winning MLB team.

Include possible out-of-the-box variables such as:
i) ownership changes
ii) funds injections
iii) change of home venues or cities
iv) MLB rule changes, global events (e.g. war)
v) climate
vi) conditions of state and national governance,
and any other variables that is of value, no matter how remote, that may affect the statistics of the game during those periods.

Moneyball Analytics – Appendix

ChatGPT Prompt 1 of 2

Hi ChatGPT, for this query, you are an experience MLB statistician and data engineer.

Using the data sources added to this chat ONLY, please provide the variables that contribute to each of the following skills evaluation metrics in MLB.

Exclude the skills evaluation that cannot be derived from the attached data sources.

A) Offensive Evaluation Metrics
1. On-Base Percentage (OBP)
2. Slugging Percentage (SLG)
3. OPS (On-Base Plus Slugging)
4. Launch Angle and Exit Velocity
5. Runs Created
6. Plate Appearances
7. Runs Batted In (RBI)
8. Strikeout rate
9. Pitches per plate appearance

B) Pitching Evaluation Metrics
1. Earned Run Average (ERA)
2. Saves (SV)
3. WHIP (Walks plus Hits per Innings Pitched)
4. Spin Rate and Velocity
5. Hits Allowed and Innings Pitched

C) Defensive and Specialized Metrics
1. Fielding Percentage
2. Range Factor
3. Outfielder Route Efficiency
4. Pitch Framing
5. Catch Probability

D) Comprehensive Value Metrics
1. WAR (Wins Above Replacement)
2. TPR (Total Player Rating)
3. Pythagorean Winning Percentage
4. Adjusted OPS+

Moneyball Analytics – Appendix

ChatGPT Prompt 2 of 2

Convert them into feature engineering pipelines (Python) to fulfill the below objective:

- Predict the number of games a team will win in a season based on their performance statistics.

Your model should be able to generalize across different eras of baseball.

Evaluation Metric:
Predictions are evaluated using Mean Absolute Error (MAE), which measures the average absolute difference between the predicted wins and actual wins. The lower the MAE, the better the model's performance.

Prediction files should be in the format of the attached 'sample_submission.csv'.

Construct 3 separate ML models below, to run the above prediction in separate sections within one .ipynb notebook file.
1. auto-sklearn (https://automl.github.io/auto-sklearn/master/)
2. autoML (run 10 random models one by one and select the best one)
3. autoML ensemble (random select 5 best models to construct to run the predictions)

Please include clear instructions in each model to show users which hyperparameters can be adjusted to achieve desired prediction outcome.

All 3 models should output a prediction csv based on the attached 'sample_submission.csv'.

IMPORTANT: Ensure the models can be executed independently based on environment.yml requirements.

Each output prediction file should be named separately in the format 'prediction_submission_*.csv' where * refers to model 1, 2 or 3 as above.

Finally, create a download link for this .ipynb notebook file available. Do consult me if there are any unclear instructions.

Moneyball Analytics – Appendix

Reverse Engineered Prompt 1 of 2

You are an experienced MLB statistician, data scientist, and ML engineer.

I am working with structured MLB team-season data (train + prediction datasets). Your task is to design a complete, production-ready machine learning pipeline to predict the number of wins (W) for each team-season.

OBJECTIVE
Predict team wins (W) using historical performance statistics.

Evaluation metric:
- Mean Absolute Error (MAE)

The model must generalize across different baseball eras (i.e., avoid temporal leakage).

DATA CONSTRAINTS
- Use ONLY the provided tabular datasets (no external)
- Data includes team-level batting, pitching, and fielding stats
- Assume typical Lahman-style schema (e.g., H, BB, AB, HR, ER, IPouts, R, RA, etc.)
- Prediction dataset has same schema except for target

FEATURE ENGINEERING REQUIREMENTS
Derive only metrics that can be computed from available columns:

Include:
- OBP (simplified), SLG, OPS
- Runs Created (basic)
- PA (approximation)
- Strikeout rate
- ERA (recomputed)
- WHIP
- Pythagorean win %
- Any useful ratio-based features (runs/game, etc.)

Exclude:
- Statcast metrics (launch angle, exit velocity)
- WAR, OPS+, advanced tracking metrics

Also:
- Create era-based features using yearID
- Ensure numerical stability (safe division)

DATA PREPROCESSING
You MUST:
- Convert boolean columns to integers (0/1)
- Handle pandas nullable boolean dtype
- Convert categorical columns using OneHotEncoder
- Impute missing values:
  - numeric → median
  - categorical → most frequent
- Standardize numeric features
- Ensure prediction dataset columns align with training

IMPORTANT:
- Avoid dtype issues that break sklearn (e.g., bool in SimpleImputer)

VALIDATION STRATEGY
Use:
- GroupKFold grouped by yearID

Reason:
- Prevent leakage across seasons
- Ensure generalization across eras

Moneyball Analytics – Appendix

Reverse Engineered Prompt 2 of 2

MODEL REQUIREMENTS - Build 3 separate models:

MODEL 1: auto-sklearn
- Use AutoSklearnRegressor
- Make it robust to:
  - temp folder conflicts
  - unsupported parameters (e.g., output_folder in older versions)
- Allow hyperparameters:
  - time_left_for_this_task
  - per_run_time_limit
  - ensemble_size
  - CV folds

MODEL 2: Custom AutoML
- Run N random trials (default ~18)
- Each trial:
  - randomly sample model + hyperparameters
  - evaluate via GroupKFold MAE
- Supported models:
  - Ridge
  - ElasticNet
  - RandomForest
  - ExtraTrees
  - GradientBoosting
  - HistGradientBoosting
  - LightGBM

IMPORTANT:
- Fix max_features bug:
  - numeric values MUST be float (e.g., 0.5, not "0.5")
- Add trial-level exception handling (do not crash loop)

MODEL 3: Ensemble
- Run larger trial pool (~24)
- Select top K models (default 5)
- Prefer diversity across model families
- Combine predictions using:
  - inverse MAE weighting

OUTPUT REQUIREMENTS
Each model must output:
- CSV file matching sample_submission.csv format

File names:
- prediction_submission_1.csv
- prediction_submission_2.csv
- prediction_submission_3.csv

NOTEBOOK REQUIREMENTS
Provide a Jupyter notebook structured into clear cells:
1. Imports
2. Config
3. Data loading
4. Feature engineering
5. Preprocessing utilities
6. Model 2 (AutoML)
7. Model 2 output
8. Model 3 (ensemble)
9. Model 3 output
10. Summary

ROBUSTNESS REQUIREMENTS
Ensure the code:
- handles dtype issues (bool, category, object)
- avoids sklearn parameter errors
- continues execution if a trial fails
- aligns train/test feature columns safely
- is reproducible (random_state control)

HYPERPARAMETER GUIDANCE
Provide recommended defaults for:
- runtime vs accuracy tradeoff
- trial counts
- CV folds
- model-specific parameter ranges

ENVIRONMENT AWARENESS
Assume:
- scikit-learn environment (no bleeding-edge APIs)
- auto-sklearn version may be older (no output_folder support)

DELIVERABLE
1. Complete notebook code (cell-by-cell)
2. No missing dependencies
3. No runtime-breaking bugs
4. Ready to copy-paste and run

Focus on:
- Correctness, Robustness
- Reproducibility, practical usability